

FACE DETECTION – EMOTION DETECTOR

What is Face Detection?

Face Detection means helping the computer find a face in a picture or camera view. PictoBlox can analyse the camera image and check whether a face is present.

The important block is: **analyse image from camera**

Then the program checks: **get faces > 0**

This means: "Is there at least one face in the camera?"

If the answer is **Yes**, the program continues.

What is Expression Detection?

After finding a face, PictoBlox can check the facial expression.

For example:

expression of face 1 = happy

means: PictoBlox thinks that Face 1 has a happy expression.

The program checks the expressions one by one.

Add the Face Detection Extension

In PictoBlox:

Step 1: Open your project.

Step 2: Find the **Face Detection** extension.

Step 3: Add the extension.

You should now see the **Face Detection** blocks.

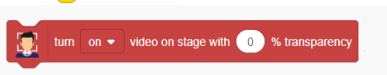
Create the Main Program

Select the main character sprite.

Start with:



Then:



This turns on the computer camera.

Next:

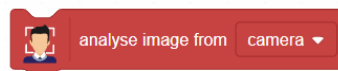


This means:

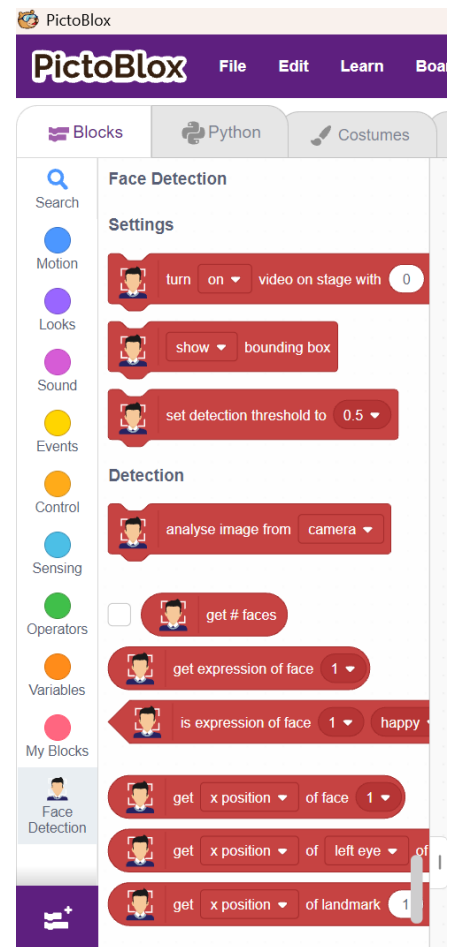
Keep checking the camera again and again.

Check for a Face

Inside forever, use:



Then check:



get faces > 0

What does get faces > 0 mean?

If the camera sees:

0 faces

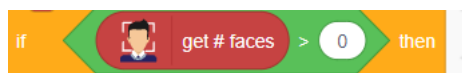
there is no face.

If it sees:

1 face

or more, a face has been detected.

So:



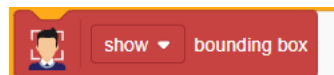
means:

"If a face is found, continue."

Show the Face Detection Box

Inside the condition, use:

show bounding box



The bounding box helps show where the detected face is.

Check the Expression

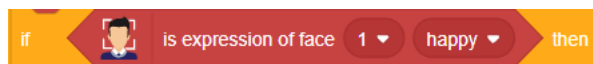
Now we want PictoBlox to check the expression of **Face 1**.

The program uses:

expression of face 1

It then compares the result with different expressions.

For example:

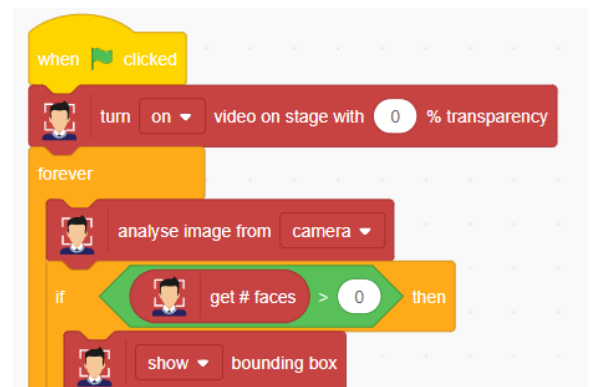


If PictoBlox detects **happy**, it sends the message:

Happy

It also makes the main character say:

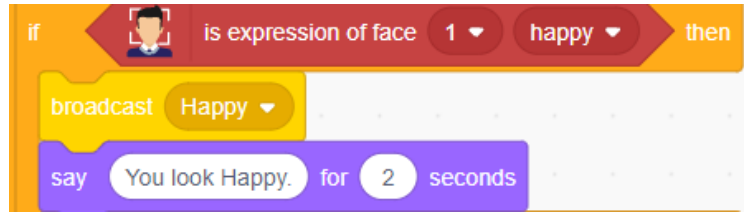
You look HAPPY.



Add the Expression Checks

if expression = happy

broadcast Happy



Broadcast is a message sent from one sprite to another.

For example:

broadcast Happy

means: "Show the Happy character."

The character sprite is waiting for that message.

It uses: when I receive Happy

Then it can show itself.

Character Sprites

Our project has different character sprites for the detected expressions.

The sprites shown in the project are:

- Sad
- Neutral
- Angry
- Surprised
- Scared
- Disgusted

The main character is also used to detect the face.

Scratch script editor showing a sequence of 'when I receive' and 'hide' blocks for various emotions. The script is as follows:

- when clicked → hide
- when I receive Angry → hide
- when I receive Sad → show
- when I receive disgusted → hide
- when I receive Normal → hide
- when I receive Happy → hide
- when I receive Scared → hide
- when I receive Surprised → hide

Backpack

Scratch stage showing a sad cat sprite and a robot. The sprite is named 'Sad' and is positioned at x=71, y=-105. The robot is positioned at x=71, y=-105.

Sprite: Sad

Size: 100

Direction: 90

Stage

Backdrops: 1

Scratch script editor showing a sequence of 'when I receive' and 'hide' blocks for various emotions. The script is as follows:

- when clicked → hide
- when I receive Normal → show
- when I receive Angry → hide
- when I receive disgusted → hide
- when I receive Scared → hide
- when I receive Sad → hide
- when I receive Surprised → hide
- when I receive Happy → hide

Backpack

Scratch stage showing a neutral cat sprite and a robot. The sprite is named 'Neutral' and is positioned at x=76, y=-109. The robot is positioned at x=76, y=-109.

Sprite: Neutral

Size: 100

Direction: 90

Stage

Backdrops: 1

Scratch script editor showing a sequence of events for an Angry sprite:

- when clicked: hide
- when I receive Angry: show
- when I receive disgusted: hide
- when I receive Normal: hide
- when I receive Sad: hide
- when I receive Sad: hide
- when I receive Surprised: hide

Backpack

Scratch stage showing an Angry emoji and a robot sprite. The sprite is named Isaac.

Sprite: Angry

Size: 100

Direction: 90

Stage

Backdrops: 1

Scratch script editor showing a sequence of events for a Surprised sprite:

- when clicked: hide
- when I receive Surprised: show
- when I receive Normal: hide
- when I receive Angry: hide
- when I receive Scared: hide
- when I receive Sad: hide
- when I receive disgusted: hide

Backpack

Scratch stage showing a Surprised emoji and a robot sprite. The sprite is named Isaac.

Sprite: Surprised

Size: 100

Direction: 90

Stage

Backdrops: 1

```

when clicked
hide
when I receive Scared
show
when I receive Angry
hide
when I receive Happy
hide
when I receive Normal
hide
when I receive Sad
hide
when I receive Surprised
hide
when I receive Disgusted
hide
when I receive Angry
hide

```

Backpack

Sprite: Scared

Size: 100

Direction: 90

Stage

Backdrop: 1

```

when clicked
hide
when I receive Disgusted
show
when I receive Angry
hide
when I receive Happy
hide
when I receive Normal
hide
when I receive Sad
hide
when I receive Scared
hide
when I receive Surprised
hide

```

Backpack

Sprite: Disgusted

Size: 100

Direction: 90

Stage

Backdrop: 1

How the Whole Program Works

Imagine you are standing in front of the camera.

PictoBlox does this:

Camera ON



Look at camera



Is there a face?



YES



Check expression



Expression detected



Broadcast a message



Matching character receives message



Character appears

